

Day – 8 VLAN Trunking and Port Accessing.

Task – 1 Connect switches through cables.

- Step 1: Configure Switch1 :- Open Switch1, go to the CLI, and enter the following commands to set FastEthernet0/1 as an 802.1Q trunk port:
 - enable
 - configure terminal
 - interface FastEthernet0/1
 - switchport trunk encapsulation dot1q
 - switchport mode trunk
 - End
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- Step 2: Configure Switch2 :- Open Switch2, go to the CLI, and run the exact same configuration on its FastEthernet0/1 port:
 - enable
 - configure terminal
 - interface FastEthernet0/1
 - switchport trunk encapsulation dot1q
 - switchport mode trunk
 - End
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- Step 3: Verify the Trunk Link :- Once the link indicator lights turn green, verify that the trunk is successfully up and operating on 802.1Q by running this command in privileged EXEC mode on either switch:
 - show interfaces trunk

Task – 2 Create three VLANs to both switches.

- Step 1: Create VLANs on Switch1: Open Switch1, go to the CLI, and run the following commands:
 - configure terminal
 - vlan 10
 - name Engineering
 - vlan 20
 - name Marketing
 - vlan 30

- name HR
- Exit
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- Step 2: Assign Access Ports on Switch1: Still on Switch1, assign ports FastEthernet0/2, FastEthernet0/3, and FastEthernet0/4 to their respective VLANs:
- configure terminal
- interface FastEthernet0/2
- switchport mode access
- switchport access vlan 10
- interface FastEthernet0/3
- switchport mode access
- switchport access vlan 20
- interface FastEthernet0/4
- switchport mode access
- switchport access vlan 30
- End
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- Step 3: Create VLANs on Switch2: Open Switch2, go to the CLI, and create the exact same VLAN database so that frames coming across the trunk link can be properly recognized:
- configure terminal
- vlan 10
- name Engineering
- vlan 20
- name Marketing
- vlan 30
- name HR
- end
- Verification
- You can verify your configuration on either switch by running:
- show vlan brief

Task – 3 Verify the VLANs is not communicate to each other.

- Step 1: Connect the PCs to Switch1
- PC-VLAN10: Connect to Switch1 port FastEthernet0/2 (assigned to VLAN 10).
- PC-VLAN20: Connect to Switch1 port FastEthernet0/3 (assigned to VLAN 20).
- PC-VLAN30: Connect to Switch1 port FastEthernet0/4 (assigned to VLAN 30).
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- Step 2: Assign IP Addresses (Same Subnet per VLAN, or separate subnets)
- For testing intra-VLAN communication, let's assume you place them on the same or different subnets. To test intra-VLAN communication correctly, you need at least two PCs in the *same* VLAN.
- Let's add a second PC to VLAN 10 to test same-VLAN communication, while keeping our primary three PCs in VLANs 10, 20, and 30 to test isolation.
- IP Configuration Example:
- PC1 (VLAN 10): IP 192.168.10.10, Subnet Mask 255.255.255.0
- PC2 (VLAN 10 - Second PC): IP 192.168.10.20, Subnet Mask 255.255.255.0
- PC3 (VLAN 20): IP 192.168.20.10, Subnet Mask 255.255.255.0
- PC4 (VLAN 30): IP 192.168.30.10, Subnet Mask 255.255.255.0
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- Step 3: Verify Isolation (Different VLANs Cannot Communicate)
- Go to PC1 (VLAN 10), open the Command Prompt, and try to ping PC3 (VLAN 20) at 192.168.20.10 or PC4 (VLAN 30) at 192.168.30.10.
- ping 192.168.20.10

Task – 4 Changing trunk link between switch 1 and switch 2.

- Steps Configuration Attempt
- Open Switch1, enter the interface configuration mode for your trunk port (FastEthernet0/1), and run:
- enable
- configure terminal
- interface FastEthernet0/1
- switchport trunk encapsulation isl
- switchport mode trunk
- end
- Repeat the exact same commands on Switch2's trunk port (FastEthernet0/1).

Task – 5 Changing the port of Switch to VLANs.

- The Impact on Connectivity Untagged vs. Tagged Traffic Mismatch:
- PCs are end devices, meaning they generate and expect untagged Ethernet frames. They do not understand or insert 802.1Q VLAN tags.
- Trunk ports are designed to carry traffic for multiple VLANs by adding 802.1Q tags to frames traveling between switches.
- The Native VLAN Fallback:

- Because the PC sends untagged frames into a trunk port, the switch treats this incoming traffic as part of the Native VLAN (which defaults to VLAN 1 on Cisco switches).
- Even though you intended for that port to belong to VLAN 20, the switch ignores that intent because a trunk port strips or assigns tags based on trunking rules rather than static access assignments.
- Total Loss of Communication with VLAN 20:
- The PC will find itself stranded on VLAN 1 instead of VLAN 20.
- As a result, it will fail to communicate with other devices in VLAN 20, as its traffic is either dropped or misrouted into the wrong logical broadcast domain.
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